

VACCINES, MONOCLONAL ANTIBODIES AND BIOPHARMA

IMMUNE-OBJECT RECOGNITION MAP → ACTIVE PASSIVE ANTIVIRAL TRIAD → ADAPTIVE IMMUNITY PROCESS RAIL → VACCINE PLATFORM TAXONOMY → GENETIC-PLATFORM CELLULAR MAP → MONOCLONAL ANTIBODY PRODUCTION CHAIN

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • source generation g3 and all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

00

STAGE 00 IMMUNE-OBJECT RECOGNITION MAP

IMMUNE-OBJECT RECOGNITION MAP CONTAINS OR PRESENTS ANTIGEN SPECIFIC RECOGNISED PART OF ANTIGEN B-CELL RECEPTOR BINDS EPITOPE T-CELL RECEPTOR RECOGNISES PRESENTED ANTIGEN FRAGMENT ANTIGEN CAPABLE OF PROVOKING AN IMMUNE RESPONSE

PATHOGEN / PRODUCT → contains or presents ANTIGEN

EPITOPE → specific recognised part of antigen

B-CELL RECEPTOR / ANTIBODY → binds epitope

T-CELL RECEPTOR → recognises presented antigen fragment

CONCEPT / ARCHITECTURE

- PATHOGEN / PRODUCT → contains or presents ANTIGEN
- EPITOPE → specific recognised part of antigen

MECHANISM / APPLICATION

- B-CELL RECEPTOR / ANTIBODY → binds epitope
- T-CELL RECEPTOR → recognises presented antigen fragment

READINESS / STATUS LIMIT

- IMMUNOGEN → antigen capable of provoking an immune response
- PATHOGEN / PRODUCT → contains or presents ANTIGEN

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: IMMUNOGEN → antigen capable of provoking an immune response.

01

STAGE 01 ACTIVE PASSIVE ANTIVIRAL TRIAD

ACTIVE PASSIVE ANTIVIRAL TRIAD ENDOGENOUS RESPONSE MEMORY MAB READY-MADE ANTIBODY DIRECT BINDING TEMPORARY EFFECT REPLICATION-STEP INHIBITION PHARMACOLOGICAL ACTION

SYSTEM / DEVICE / INSTITUTION

VACCINE → antigen/code → endogenous response → memory mAb → ready-made antibody → direct binding → temporary effect
ANTIVIRAL → replication-step inhibition → pharmacological action

DECISIVE TECHNICAL DISTINCTION

ONSET / DURATION / PURPOSE → compare, never merge

STATUS / UNIT / EVIDENCE LIMIT

ANTIVIRAL → replication-step inhibition → pharmacological action

SYSTEM / DEVICE / INSTITUTION

- VACCINE → antigen/code → endogenous response → memory mAb → ready-made antibody → direct binding → temporary effect
- ANTIVIRAL → replication-step inhibition → pharmacological action

DECISIVE TECHNICAL DISTINCTION

- ONSET / DURATION / PURPOSE → compare, never merge
- VACCINE → antigen/code → endogenous response → memory mAb → ready-made antibody → direct binding → temporary effect

STATUS / UNIT / EVIDENCE LIMIT

- ANTIVIRAL → replication-step inhibition → pharmacological action
- ONSET / DURATION / PURPOSE → compare, never merge

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: ONSET / DURATION / PURPOSE → compare, never merge.

02

STAGE 02 ADAPTIVE IMMUNITY PROCESS RAIL

ADAPTIVE IMMUNITY PROCESS RAIL ANTIGEN PRESENTATION LYMPHOCYTE ACTIVATION B CELL PLASMA CELL T CELL MEMORY POPULATIONS HELPER T CELL

ANTIGEN PRESENTATION → lymphocyte activation

B CELL → plasma cell → antibodies

B / T CELL → memory populations

HELPER T CELL → coordination; CYTOTOXIC T CELL → cellular killing

RULE → antibody titre is not the entire immune response

STARTING CONDITION / INPUT

- ANTIGEN PRESENTATION → lymphocyte activation
- B CELL → plasma cell → antibodies

SCIENTIFIC OR ENGINEERING MECHANISM

- B / T CELL → memory populations
- HELPER T CELL → coordination; CYTOTOXIC T CELL → cellular killing

OUTPUT / FAILURE BOUNDARY

- RULE → antibody titre is not the entire immune response
- ANTIGEN PRESENTATION → lymphocyte activation

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → antibody titre is not the entire immune response.

03

STAGE 03 VACCINE PLATFORM TAXONOMY

VACCINE PLATFORM TAXONOMY WHOLE PATHOGEN LIVE ATTENUATED SELECTED MATERIAL GENETIC CODE VIRAL VECTOR ANTIGEN FORM CELL LOCATION

WHOLE PATHOGEN → live attenuated

inactivated

SELECTED MATERIAL → subunit

toxoid

VLP

conjugate

- ANTIGEN PRESENTATION → lymphocyte activation
- B CELL → plasma cell → antibodies

- B / T CELL → memory populations
- HELPER T CELL → coordination; CYTOTOXIC T CELL → cellular killing

- RULE → antibody titre is not the entire immune response
- ANTIGEN PRESENTATION → lymphocyte activation

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → antibody titre is not the entire immune response.

03

STAGE 03VACCINE PLATFORM TAXONOMY

VACCINE PLATFORM TAXONOMYWHOLE PATHOGENLIVE ATTENUATEDSELECTED MATERIALGENETIC CODEVIRAL VECTORANTIGEN FORMCELL LOCATION

WHOLE PATHOGEN → live attenuated	inactivated		
SELECTED MATERIAL → subunit	toxoid	VLP	conjugate
GENETIC CODE → viral vector	mRNA	DNA	
COMPARE → antigen form	replication	cell location	manufacture

SYSTEM / DEVICE / INSTITUTION

- WHOLE PATHOGEN → live attenuated
- SELECTED MATERIAL → subunit

DECISIVE TECHNICAL DISTINCTION

- GENETIC CODE → viral vector
- COMPARE → antigen form

STATUS / UNIT / EVIDENCE LIMIT

- cell location
- DO NOT IMPORT → product approval, efficacy or present availability

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: DO NOT IMPORT → product approval, efficacy or present availability.

04

STAGE 04GENETIC-PLATFORM CELLULAR MAP

GENETIC-PLATFORM CELLULAR MAPVIRAL VECTORENGINEERED CARRIER DELIVERS ANTIGEN CODE MRNA + LNPANTIGEN EXPRESSIONB/T RESPONSEACTIVE IMMUNITY DOES NOT MEAN IDENTICAL DELIVERY

VIRAL VECTOR → engineered carrier delivers antigen code mRNA + LNP → CYTOPLASM → translation

DNA → NUCLEUS → transcription → mRNA

ANTIGEN EXPRESSION → B/T response → memory

TRAP → active immunity does not mean identical delivery

CONCEPT / ARCHITECTURE

- VIRAL VECTOR → engineered carrier delivers antigen code mRNA + LNP → CYTOPLASM → translation
- DNA → NUCLEUS → transcription → mRNA

MECHANISM / APPLICATION

- ANTIGEN EXPRESSION → B/T response → memory
- TRAP → active immunity does not mean identical delivery

READINESS / STATUS LIMIT

- VIRAL VECTOR → engineered carrier delivers antigen code mRNA + LNP → CYTOPLASM → translation
- DNA → NUCLEUS → transcription → mRNA

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRAP → active immunity does not mean identical delivery.

05

STAGE 05MONOCLONAL ANTIBODY PRODUCTION CHAIN

MONOCLONAL ANTIBODY PRODUCTION CHAINB CELL + MYELOMAHYBRIDOMA CLONEOR SEQUENCE DISCOVERYANTIBODY ENGINEERINGRECOMBINANT MAMMALIAN CELL LINETARGET BINDINGPROVEN CLINICAL BENEFIT

B CELL + MYELOMA → HYBRIDOMA CLONE

OR SEQUENCE DISCOVERY → ANTIBODY ENGINEERING

RECOMBINANT MAMMALIAN CELL LINE → EXPRESSION

HARVEST → PURIFICATION → FORMULATION → FILL-FINISH

TARGET BINDING != PROVEN CLINICAL BENEFIT

STARTING CONDITION / INPUT

- B CELL + MYELOMA → HYBRIDOMA CLONE
- OR SEQUENCE DISCOVERY → ANTIBODY ENGINEERING

SCIENTIFIC OR ENGINEERING MECHANISM

- RECOMBINANT MAMMALIAN CELL LINE → EXPRESSION
- HARVEST → PURIFICATION → FORMULATION → FILL-FINISH

OUTPUT / FAILURE BOUNDARY

- TARGET BINDING != PROVEN CLINICAL BENEFIT
- B CELL + MYELOMA → HYBRIDOMA CLONE

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: HARVEST → PURIFICATION → FORMULATION → FILL-FINISH; TARGET BINDING != PROVEN CLINICAL BENEFIT.

06

STAGE 06BIOPHARMA MANUFACTURING QUALITY SYSTEM

BIOPHARMA MANUFACTURING QUALITY SYSTEMWORKING CELL BANK + RAW MATERIAL CONTROLSTERILE PRODUCT AND PACKAGINGMASTER / WORKING CELL BANK + RAW MATERIAL CONTROL

MASTER / WORKING CELL BANK + RAW MATERIAL CONTROL

UPSTREAM → culture / fermentation / bioreactor

DOWNSTREAM → capture / purification / clearance

FILL-FINISH → sterile product and packaging

CONCEPT / ARCHITECTURE

- MASTER / WORKING CELL BANK + RAW MATERIAL CONTROL

MECHANISM / APPLICATION

- DOWNSTREAM → capture / purification / clearance

READINESS / STATUS LIMIT

- RELEASE → identity / purity / potency / sterility / consistency

06

STAGE 06 BIOPHARMA MANUFACTURING QUALITY SYSTEM

BIOPHARMA MANUFACTURING QUALITY SYSTEM

WORKING CELL BANK + RAW MATERIAL CONTROL

STERILE PRODUCT AND PACKAGING

MASTER / WORKING CELL BANK + RAW MATERIAL CONTROL

MASTER / WORKING CELL BANK + RAW MATERIAL CONTROL

UPSTREAM → culture / fermentation / bioreactor

DOWNSTREAM → capture / purification / clearance

FILL-FINISH → sterile product and packaging

CONCEPT / ARCHITECTURE

- MASTER / WORKING CELL BANK + RAW MATERIAL CONTROL
- UPSTREAM → culture / fermentation / bioreactor

MECHANISM / APPLICATION

- DOWNSTREAM → capture / purification / clearance
- FILL-FINISH → sterile product and packaging

READINESS / STATUS LIMIT

- RELEASE → identity / purity / potency / sterility / consistency
- MASTER / WORKING CELL BANK + RAW MATERIAL CONTROL

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RELEASE → identity / purity / potency / sterility / consistency.

07

STAGE 07 COLD-CHAIN INTEGRITY LOOP

COLD-CHAIN INTEGRITY LOOP

VALIDATED STORAGE

MONITORED TRANSPORT

DELIVERY POINT

DATA LOGGER

DEVIATION DETECTION

POWER, EQUIPMENT, ROUTE AND STOCK RESPONSE

INTEGRITY ENABLES ACCESS; IT DOES NOT PROVE UPTAKE

STARTING CONDITION / INPUT

- VALIDATED STORAGE → monitored transport → delivery point
- DATA LOGGER / INDICATOR → deviation detection

SCIENTIFIC OR ENGINEERING MECHANISM

- EXCURSION → quarantine / assessment / disposition
- CONTINGENCY → power, equipment, route and stock response

OUTPUT / FAILURE BOUNDARY

- OUTCOME → integrity enables access; it does not prove uptake
- VALIDATED STORAGE → monitored transport → delivery point

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: OUTCOME → integrity enables access; it does not prove uptake.

08

STAGE 08 CLINICAL AND REGULATORY LADDER

CLINICAL AND REGULATORY LADDER

LABORATORY AND ANIMAL EVIDENCE

PHASE I

INITIAL SAFETY

PHASE II

EXPANDED DOSE, SAFETY AND RESPONSE EVIDENCE

PHASE III

LARGER-POPULATION BENEFIT AND SAFETY

PRECLINICAL → laboratory and animal evidence

PHASE I → initial safety / dose

PHASE II → expanded dose, safety and response evidence

PHASE III → larger-population benefit and safety

STARTING CONDITION / INPUT

- PRECLINICAL → laboratory and animal evidence
- PHASE I → initial safety / dose

SCIENTIFIC OR ENGINEERING MECHANISM

- PHASE II → expanded dose, safety and response evidence
- PHASE III → larger-population benefit and safety

OUTPUT / FAILURE BOUNDARY

- AUTHORISATION → MANUFACTURE / RELEASE → PHASE IV SURVEILLANCE
- PRECLINICAL → laboratory and animal evidence

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: AUTHORISATION → MANUFACTURE / RELEASE → PHASE IV SURVEILLANCE.

09

STAGE 09 INDIAN INSTITUTION ROUTER

INDIAN INSTITUTION ROUTER

BIOTECH POLICY, MISSIONS AND RESEARCH SUPPORT

TRANSLATION, ENTERPRISE AND MISSION IMPLEMENTATION

BIOMEDICAL RESEARCH AND EVIDENCE

TRIALS AND NEW-DRUG

BIOLOGIC REGULATION

PHARMACOVIGILANCE COORDINATION AND SAFETY LEARNING

MUST REMEMBER

DBT → biotech policy, missions and research support

BIRAC → translation, enterprise and mission implementation

ICMR → biomedical research and evidence

CDSCO / DCGI → trials and new-drug / biologic regulation

CONCEPT / ARCHITECTURE

- DBT → biotech policy, missions and research support
- BIRAC → translation, enterprise and mission implementation

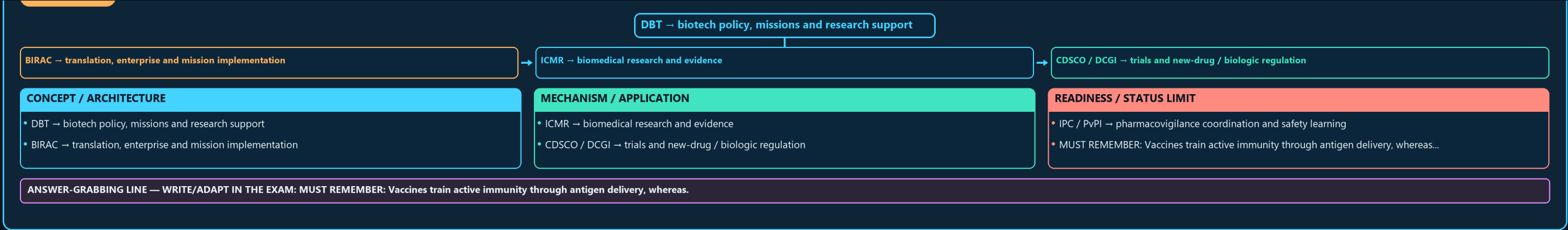
MECHANISM / APPLICATION

- ICMR → biomedical research and evidence
- CDSCO / DCGI → trials and new-drug / biologic regulation

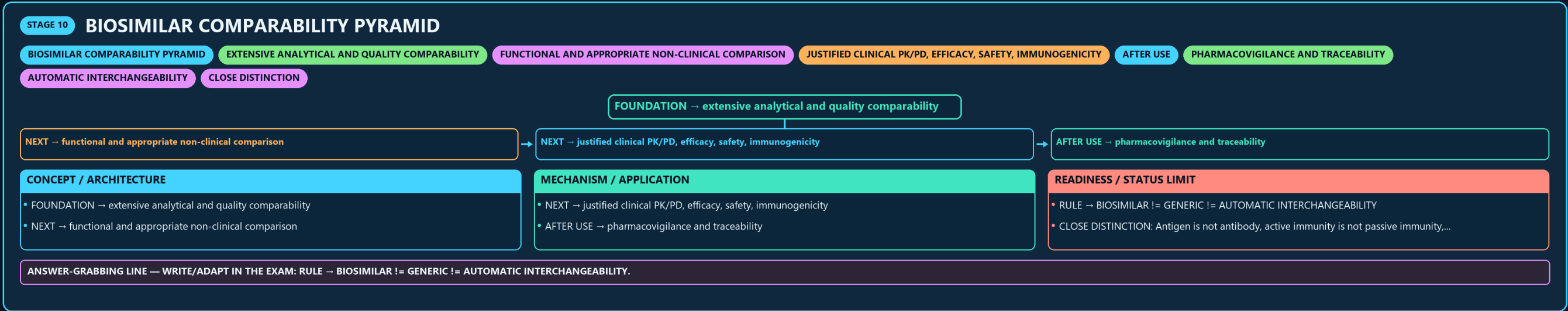
READINESS / STATUS LIMIT

- IPC / PvPI → pharmacovigilance coordination and safety learning
- MUST REMEMBER: Vaccines train active immunity through antigen delivery, whereas...

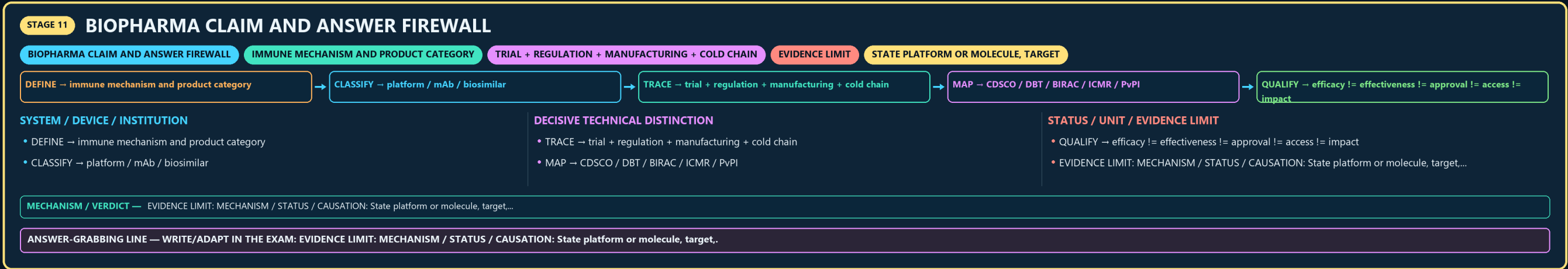
ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MUST REMEMBER: Vaccines train active immunity through antigen delivery, whereas.



10



11



E

